

# ReGo Travel Management System - Documentation Package

## Complete Implementation Guide for Hackathon

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### What's Included

This package contains everything you need to build the ReGo Corporate Travel Management System from scratch. You have **3 comprehensive documents** that cover every aspect of development.

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### Document Overview

#### 1. ReGo\_Complete\_Technical\_Documentation.md (63 KB)

**Your main reference document** - This is the heart of your project documentation.

#### Contains:

-  Complete system architecture
-  Full database schema with all tables
-  All API endpoint specifications
-  Frontend component structure
-  Authentication & security implementation
-  AI & OCR integration details
-  Complete workflows and user journeys
-  5-week development roadmap
-  Team assignment and responsibilities
-  Testing strategy
-  Deployment guide
-  Edge cases and additional features

#### When to use:

- Planning system architecture
- Designing database schema
- Creating API endpoints

- Understanding user workflows
  - Planning sprints and milestones
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## 2. ReGo\_Visual\_Diagram\_References.md (25 KB)

**Your diagramming guide** - Create professional diagrams for presentations.

### Contains:

-  Database ER diagram (ready to copy to dbdiagram.io)
-  System architecture diagram layout
-  Travel request workflow flowchart
-  Document verification flow
-  Approval chain decision logic
-  Frontend component hierarchy
-  API flow diagrams
-  Deployment architecture
-  User journey maps
-  Color schemes and design tips

### When to use:

- Creating presentations
- Explaining system to stakeholders
- Team discussions and planning
- Documentation for investors
- Weekly milestone demos

### Tools recommended:

- dbdiagram.io (for database diagrams)
  - draw.io (for all other diagrams)
  - Lucidchart (alternative)
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### 3. ReGo\_Quick\_Start\_Guide.md (13 KB)

**Your Day 1 setup guide** - Get everyone coding in 30 minutes.

#### Contains:

-  Complete prerequisites checklist
-  Step-by-step project setup
-  Backend setup commands
-  Frontend setup commands
-  Database setup instructions
-  Testing procedures
-  Common commands reference
-  Troubleshooting guide
-  Code templates (ready to copy-paste)
-  First week goals
-  Pro tips for new developers

#### When to use:

- First day of development
  - Onboarding new team members
  - When starting fresh
  - Quick reference for commands
  - Troubleshooting issues
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## How to Get Started

### Step 1: Read in This Order

1. **First 30 minutes:** Read **Quick Start Guide** → Set up your environment
2. **Next 1 hour:** Read **Technical Documentation** sections 1-3 → Understand the system
3. **Next 30 minutes:** Look at **Visual Diagram References** → Create your first diagram

### Step 2: Team Meeting (Day 1)

#### Agenda:

1. Everyone presents their setup (5 min each)
2. Assign roles from Technical Documentation Section 11
3. Review Week 1 goals from Quick Start Guide
4. Create GitHub repository
5. Decide on communication tools (Slack/Discord)

### Step 3: First Sprint (Week 1)

#### Using Quick Start Guide:

- Days 1-2: Complete project setup (all team members)
- Days 3-5: Backend lead creates auth, Frontend lead creates layout
- Days 5-7: Integrate and prepare demo

**Demo Goal:** Working login system with basic dashboard

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### Weekly Milestone Checklist

#### Week 1 ✓

- ☐ Dev environments set up
- ☐ Database created and populated
- ☐ Git repository initialized
- ☐ Authentication working
- ☐ Basic UI layout complete
- ☐ **Demo:** Login + Dashboard

#### Week 2

- ☐ Travel request CRUD complete
- ☐ Document upload working
- ☐ Approval workflow functional
- ☐ Notifications sending
- ☐ **Demo:** Full request-to-approval flow

#### Week 3

- ☐ Azure Document Intelligence integrated
- ☐ OCR processing documents
- ☐ AI flight recommendations working
- ☐ Hotel recommendations functional
- ☐ **Demo:** Document verification + AI features

## Week 4

- ☐ Booking management complete
- ☐ Expense submission working
- ☐ Finance dashboard functional
- ☐ All notifications operational
- ☐ **Demo:** Complete expense tracking

## Week 5

- ☐ All tests passing
  - ☐ UI polished and responsive
  - ☐ Deployed to Azure
  - ☐ Documentation complete
  - ☐ **Demo:** Production-ready application
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## Key Features Checklist

### Core Features (Must Have)

- ☐ User authentication (login/logout)
- ☐ Multi-role support (Employee, Manager, AVP, SVP, CHRO, Travel Desk, Finance)
- ☐ Travel request creation
- ☐ Document upload
- ☐ OCR document verification (Azure Document Intelligence)
- ☐ Multi-level approval workflow (Manager → AVP → SVP → CHRO)
- ☐ Real-time notifications (Email + SMS)
- ☐ Booking management
- ☐ Expense submission
- ☐ Reimbursement processing

### AI Features (Differentiators)

- ☐ AI flight recommendations (low/mid/high cost)
- ☐ AI hotel suggestions
- ☐ Automatic document data extraction (OCR)
- ☐ Expense anomaly detection

### Nice-to-Have (If Time Permits)

- ☐ Calendar integration
- ☐ Advanced analytics dashboard
- ☐ Export reports (PDF/Excel)
- ☐ Mobile-responsive design

## Technology Stack Summary

| Component | Technology                  | Why                                      |
|-----------|-----------------------------|------------------------------------------|
| Frontend  | React.js + Material-UI      | Modern, component-based, beautiful UI    |
| Backend   | Node.js + Express           | JavaScript everywhere, easy to learn     |
| Database  | PostgreSQL                  | Robust, supports JSONB for flexible data |
| ORM       | Prisma                      | Type-safe, easy migrations               |
| Cloud     | Azure                       | Free credits, all-in-one platform        |
| OCR       | Azure Document Intelligence | Best-in-class document processing        |
| State     | Redux Toolkit               | Predictable state management             |
| Auth      | JWT                         | Stateless, scalable authentication       |

## Support & Resources

### When You're Stuck

1. **Check the docs** you've been given
2. **Google the error** message
3. **Ask Claude or ChatGPT** for code help
4. **Check official documentation:**
  - React: <https://react.dev>
  - Node.js: <https://nodejs.org/docs>
  - Prisma: <https://www.prisma.io/docs>
  - Material-UI: <https://mui.com>
5. **Ask your team** - that's what they're there for!

### Useful Commands Quick Reference

```
bash
```

*# Start backend*

`cd backend && npm run dev`

*# Start frontend*

`cd frontend && npm start`

*# Run tests*

`npm test`

*# Check git status*

`git status`

*# Create new branch*

`git checkout -b feature/your-feature`

*# Database commands*

`psql -U postgres -d rego_db`



## Creating Your Presentation

### For Weekly Demos:

1. **Open Visual Diagram References**
  2. Go to [dbdiagram.io](https://dbdiagram.io) and create database diagram
  3. Go to [draw.io](https://draw.io) and create:
    - System architecture diagram
    - Current workflow diagram
  4. Take screenshots of working features
  5. Create slides showing:
    - What was accomplished
    - What's working (with demos)
    - Challenges faced
    - Next week's goals
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## **Pro Tips for Success**

### **Development**

1. **Commit frequently** - Small commits with clear messages
2. **Test locally first** - Before pushing to GitHub
3. **Use branches** - Never commit directly to main
4. **Code reviews** - Have Dev 5 (QA) review all PRs
5. **Follow the plan** - Stick to the weekly roadmap

### **Team Collaboration**

1. **Daily standups** - 15 minutes every morning
2. **Clear communication** - Use Slack/Discord effectively
3. **Help each other** - Pair programming when stuck
4. **Document decisions** - Keep notes of important choices
5. **Celebrate wins** - Acknowledge completed milestones

### **Presentation**

1. **Practice your demo** - Run through it multiple times
  2. **Have a backup** - Record demo video as backup
  3. **Highlight AI features** - These are your differentiators
  4. **Show the problem** - Explain what ReGo solves
  5. **Be enthusiastic** - Your excitement is contagious
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## **What Makes ReGo Special**

### **Unique Selling Points:**

1. **AI-Powered OCR** - Automatic document verification
2. **Smart Recommendations** - AI suggests best flight/hotel options
3. **Complete Transparency** - Real-time status for everyone
4. **Zero Email Chains** - All approvals in one place
5. **Fast Reimbursements** - Digital process cuts wait time
6. **Anomaly Detection** - AI flags suspicious expenses



Use these in your pitch!

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## Important Reminders

- **Week 1 Demo:** Focus on working authentication
  - **Keep It Simple:** MVP first, then add features
  - **Document As You Go:** Don't wait until the end
  - **Test Everything:** QA should test continuously
  - **Azure Credits:** Apply early to avoid delays
  - **Backup Your Work:** Push to GitHub daily
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## Final Pre-Demo Checklist

### 24 Hours Before Demo:

- ☐ All code merged to main branch
  - ☐ Application deployed (or ready to run locally)
  - ☐ Demo script prepared
  - ☐ Slides created with diagrams
  - ☐ All team members know their talking points
  - ☐ Backup demo video recorded
  - ☐ Questions anticipated and answered
  - ☐ Enthusiasm level: *100*
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## You're Ready!

You now have:

- ☒ Complete technical architecture
- ☒ Detailed database design
- ☒ All API specifications
- ☒ Frontend component structure
- ☒ Authentication system design
- ☒ AI/OCR integration plan
- ☒ 5-week development roadmap

-  Team assignments
-  Testing strategy
-  Deployment guide
-  Visual diagram references
-  Quick start guide
-  Code templates
-  Troubleshooting tips

**Everything you need to build a winning hackathon project!**

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## Questions?

If anything is unclear, you can:

1. Re-read the relevant section in the technical documentation
  2. Check the Quick Start Guide for practical steps
  3. Look at the Visual Diagram References for clarity
  4. Ask Claude or ChatGPT for code examples
  5. Discuss with your team
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**Good luck with your hackathon! You've got this! **

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*Document created: November 12, 2025*

*Version: 1.0*

*For: ReGo Travel Management System Hackathon Team*